The Paleomagnetism of Some Triassic Red Beds from Northern Rhodesia

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Abstract. During August 1963 fifteen sites were sampled in the Red Sandstone formation (upper Triassic) of the Zambezi Valley, Northern Rhodesia, covering a stratigraphic interval of more than 300 meters. Six of these sites proved to be measurable. Specimens were subjected to alternating-field and thermal demagnetization, which demonstrated that the direction of magnetization was stable. The pole position calculated from the six sites is 68°S, 50½°E ($\delta p = 5°$, $\delta m = 6½°$). This pole position is in accord with other Mesozoic results from Africa.

Introduction. During the 1961 field season oriented blocks were collected from the Triassic Red Sandstone in the valley of the Lusitu River, Northern Rhodesia. The preliminary measurement of these blocks was encouraging, so a second field trip to the area was made in August 1963 and a more extensive collection made. *Nairn* [1960] had previously reported on results obtained on three oriented blocks sent by the Northern Rhodesia Geological Survey from a site near the Lusitu River bridge.

Geology. The Red Sandstone formation [*Gair,* 1959] consists predominantly of fine-grained, well-stratified, mostly arkosic, red-colored sandstones. Limy sandstones and impure limestones are also present. This formation conformably overlies the Sandstone and Interbedded Mudstone formation. The boundary between the two is arbitrary. The relationship of the Red Sandstone formation with the overlying Batoka basalts is not clear, but an unconformity may be present. Tectonic activity seems to be confined to block faulting, which sometimes inclines the beds at angles of up to 20°, usually to the south and east. Fossils, with the exception of fossil logs, have not been found in this formation, and it is therefore impossible to correlate it exactly with the South African standard section. It is probably of Stormberg (upper Triassic) age. The formation is estimated to be 1110 m thick.

Sampling. Samples were collected at fifteen sites with a small Graham-Keiller type sampling drill [*Graham and Keiller,* 1960] using methods previously described by *Gough and Opdyke* [1963]. The sites were located in the bed of the Lusitu River, with the exception of one which was located along the Zambezi River at the 'Old Boma.' The sites usually represent only single beds, seldom exceeding 3 m in thickness, except that site 1 (Table 1) covers an interval of 15 m. At this site twelve cores 25 cm long were drilled, whereas at all the other sites only four cores were obtained. Two specimens 2.5 cm long were cut from each core, and the mean direction of magnetization for each core is regarded as a single observation. The stratigraphic interval covered exceeded 300 m in thickness.

Laboratory studies. Of the fifteen sites sampled only six had intensities of magnetization higher than the lower limit of measurement on the spinner magnetometer [*Gough,* 1964].

All specimens were subjected to alternating magnetic fields in the apparatus previously described by *McElhinny and Gough* [1963]. The specimens at only one site (no. 6.) responded to alternating fields, peak magnitude 800 gauss. No significant change in moment of any of the specimens was observed when demagnetized in this field. The alternating-field demagnetization curves are similar to that previously given by *Opdyke* [1961], which is considered to show stable behavior.

Specimens from five of the six sites were

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TABLE 1. Site Directions for the Triassic Red Sandstones of Northern Rhodesia

Site	N	Treatment	D	I	R	k	α_{95}
1	12	NRM	353.7°	−58.8°	11.739	42	6.8°
2	4	NRM	350.5	−60.5	3.996	750	3.3
3	5	NRM	354.0	−55.0	4.972	141	6.5
4	3	NRM	354.5	−49.0	2.995	385	6.3
5	4	NRM	344.0	−50.5	3.992	366	4.8
6	4	800 gauss	346.5	−59.5	3.998	1579	2.3

N is the number of samples (i.e. cores) included in the analysis.
R is the length of the resultant of the N unit vectors and D and I are the declination and inclination of its direction.
α_{95} is the semiangle of the cone of 95 per cent confidence, and $k = (N - 1)/(N - R)$ is the best estimate of Fisher's dispersion parameter.
The sites are located at 28°45′E, 16°10′S.

thermally demagnetized using the equipment and techniques previously described by *Opdyke* [1964]. The thermal demagnetization curves are shown in Figure 1. These curves show that the specimens possess 'thermally distributed' magnetic components believed to be caused by the presence of grains of hematite with different diameters [*Irving and Opdyke*, 1964]. Although the magnetic moment of the specimens dropped off steadily with increasing temperature, the direction of magnetization remained unchanged in temperatures as high as 650°C. This demonstrates that only a single direction of magnetization is present.

Pole position. Since these rocks exhibit stable behavior when subjected to both alternating-field and thermal demagnetization, it is assumed that they reflect the direction of the earth's field for a portion of the upper Triassic. The mean direction of magnetization of the Red Sandstone formation is not significantly different from the direction of the earth's present field, but it shows considerable divergence from the axial dipole field direction. However, it seems unlikely that these beds acquired their magnetization in very recent times. At two of the sites the sandstones are highly calcareous and are not red, but the rock is flecked throughout with concentrations of hematite. At these sites the acquisition of the hematite must have preceded or have been penecontemporaneous with the deposition of the calcareous cement which encloses them. Many of the beds contain calcareous nodules, some of them showing septarian structure. The rocks probably acquired their calcareous cement and the hematite flecks, which are responsible for their magnetism, on or soon after deposition. The mean direction of magnetization is similar to that found for igneous rocks of Mesozoic age which have been studied in central Africa. These igneous rocks have a wide variety of composition, and in some cases the direction of magnetization, although being close to that of the earth's present field, is opposite in sense [*Gough, Brock, Jones, and Opdyke*, 1964].

The pole position was calculated from the NRM directions at sites 1 to 5. At site 6 Fisher's dispersion parameter k [*Fisher*, 1953] improved slightly on treatment in 800 gauss, the mean moving about 3°; the mean after treatment has been used in the final calculation.

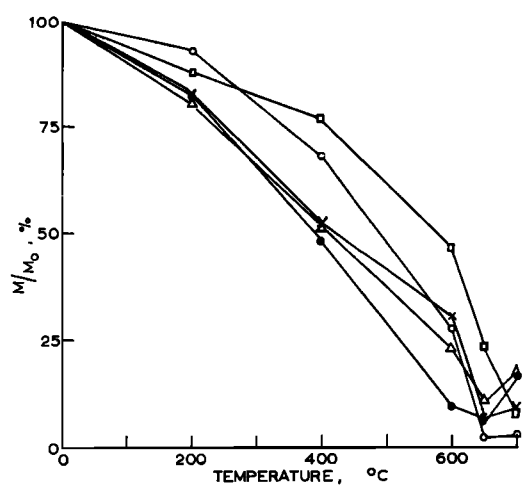


Fig. 1. Normalized thermal demagnetization curves of Red Sandstone samples.

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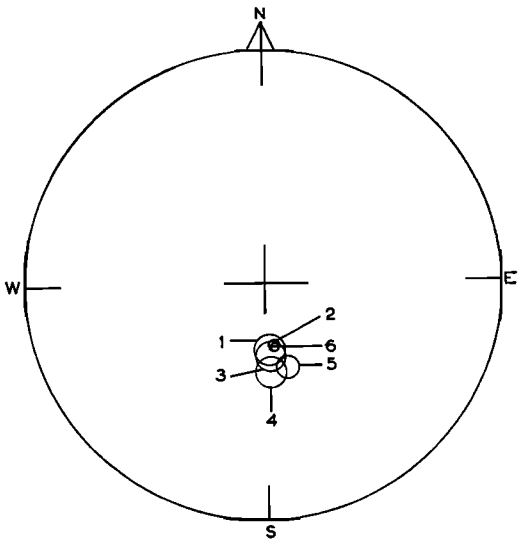


Fig. 2. Mean directions of magnetization of six sites in the Triassic Red Sandstone formation. Circles of 95 per cent confidence are plotted on a lower-hemisphere stereographic projection. The directions of magnetization are south-seeking down in all cases.

The mean direction of magnetization giving each site unit weight is $D = 350.3°$, $I = -55.6°$, $K = 208$, $\alpha_{95} = 4.6°$. On the assumption of a geocentric axial dipole field, this places the mean south magnetic pole at 68°S, 50½°E; the semiaxes of the oval of 95 per cent confidence are $\delta p = 5°$ and $\delta m = 6½°$.

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